IMPERIAL COLLEGE LONDON

EE3-10
EE9-CS3-1

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected copy

MATHEMATICS FOR SIGNALS AND SYSTEMS

Friday, 14 December 9:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions. All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Special Information for the Invigilators: none

Information for Candidates: none

The Questions

1. (a) Let V be the vector space consisting of all functions of the form

$$f(x) = \alpha \cos x + \beta \sin x.$$

Consider the linear transformation

$$L = \frac{d^2 f}{dx^2} - \frac{df}{dx} + 2f(x).$$

i. Find the matrix representing the linear mapping L with respect to the basis $\{\cos x, \sin x\}$ [2]

ii. Use the answer from part i. to find one solution to the following differential equation:

$$\frac{d^2 f}{dx^2} - \frac{df}{dx} + 2f(x) = 3\cos x - \sin x$$

[2]

(b) Show that if $\mathbf{AB} = \mathbf{0}$ then the range space of $\mathbf{B}$ is in the null space of $\mathbf{A}$. [2]

(c) Consider the following subspace of $\mathbb{R}^4$:

$$S = \text{ span} \left\{ \begin{bmatrix} 1 \\ 4 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 4 \\ -1 \end{bmatrix} \right\}.$$

Find a basis for S. [3]

Question 1 continues on the next page

(d) Find the dimension and construct a basis for the four subspaces associated with

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & -1 & 1 \end{bmatrix}.$$

[5]

(e) Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 5 \end{bmatrix}$$

i. Find the LU factorization of $\mathbf{A}$ [3]

ii. Find the determinant of $\mathbf{A}$ [3]

2. (a) The sequence x_n has support $N = 4$, that is, it is assumed to be zero for $n \neq 0, 1, ..., N-1$. This sequence is filtered with a filter h_n with unit impulse response

$$h_n = \begin{cases} 1 & \text{for } n = 0 \text{ and } n = 1 \\ 0 & \text{otherwise.} \end{cases} \tag{2a}$$

We denote the filtered sequence with $y_n = h_n * x_n$ where $*$ denotes the convolution operation.

i. The linear convolution can be written in matrix/vector form as $\mathbf{y} = \mathbf{Hx}$. Write the exact expression of the matrix $\mathbf{H}$ that describes the linear convolution with the filter h_n [3]

ii. What are the dimensions of $\mathbf{H}$? [1]

iii. Find a basis for the range and null spaces of $\mathbf{H}$ [3]

iv. Assume now that $N = 2$, i.e., x_n is assumed to be zero for $n \neq 0, 1$.

A. Write the exact expression of the matrix $\mathbf{H}$ that describes the linear convolution with the filter h_n as in Eq. (2a). [1]

B. You observe $\mathbf{y} = [1, 2, 2]^T$ which is a noisy version of $y_n = h_n * x_n$, that is, $\mathbf{y} = \mathbf{Hx} + \mathbf{e}$ for some noise $\mathbf{e}$. This means that you cannot recover $\mathbf{x}$ exactly. Find the least-squares solution to $\mathbf{y} = \mathbf{Hx}$, with $\mathbf{H}$ as in Eq. (2a) and with $\mathbf{y} = [1, 2, 2]^T$. [3]

Question 2 continues on the next page

(b) Assume now that x_n is periodic with period $N = 4$.

i. Write the exact expression of the matrix $\mathbf{H}$ that describes the circulant convolution with the filter h_n as in Eq. (2a). [3]

ii. What are the dimensions of $\mathbf{H}$? [1]

iii. Find a basis for the range and null spaces of $\mathbf{H}$. [3]

iv. Given that $y_n = h_n * x_n$ where now $*$ denotes the circulant convolution, can you always retrieve x_n from y_n? Justify your answer. [2]

3. (a) Bob and Alice play cards everyday and everyday Alice looses 1/10 of her total amount of money to Bob. At the same time Bob loses 1/4 of his total amount of money in favour of Alice.

   i. Assume that Alice had £25 at the beginning and Bob had £10, compute the total amount of money that Alice has after 2 days. [5]

   ii. Assume now that they can play an infinite number of times and that they both had £10 at the beginning, compute the total amount of money that Bob has after an *infinite* number of days. [3]

(b) Show that the eigenvectors related to two different eigenvalues of a symmetric matrix are orthogonal. [4]

(c) Consider the systems of linear equations $\mathbf{y} = \mathbf{A}\mathbf{x}$ with $\mathbf{A}$ given by:

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 0 & \alpha \end{bmatrix}.$$

   i. Find a real value of α that guarantees that the system has always at least one solution. Justify your answer. [2]

   ii. Given the α you found and assuming that $\mathbf{y} = [-2, 1]^T$, find the minimum norm solution to $\mathbf{y} = \mathbf{A}\mathbf{x}$. [2]

Question 3 continues on the next page

(d) Let $\mathbf{y}$ be the two-dimensional vector $\mathbf{y} = [0.5, 1]^T$ and let the scalar $\alpha = 1$. Find the two dimensional vector $\mathbf{x} = [x_1, x_2]^T$ that is closest in the least-squares sense to $\mathbf{y}$ among all two-dimensional vectors that have average value $\alpha = 1$. That is find the $\mathbf{x}$ that minimizes $\|\mathbf{y} - \mathbf{x}\|^2$ subject to $(x_1/2 + x_2/2) = 1$.

[4]